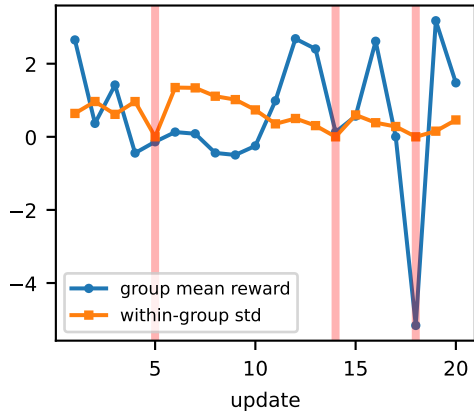
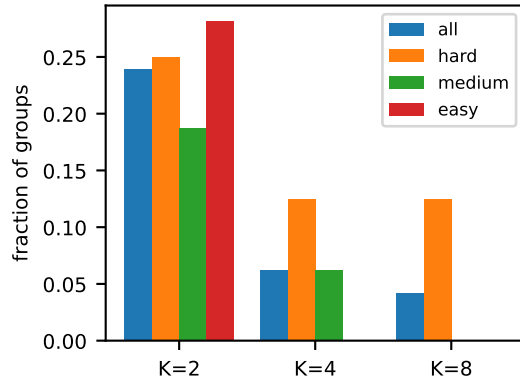


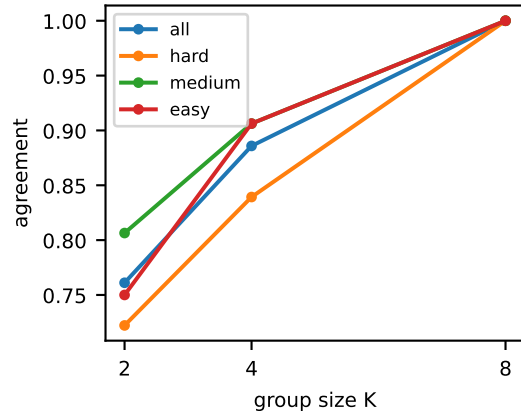
(a) Standard GRPO (red = uninformative)



(b) Low-signal groups (std<0.1)



(c) Advantage sign agreement with K=8



(d) Share of gradient weight by length

